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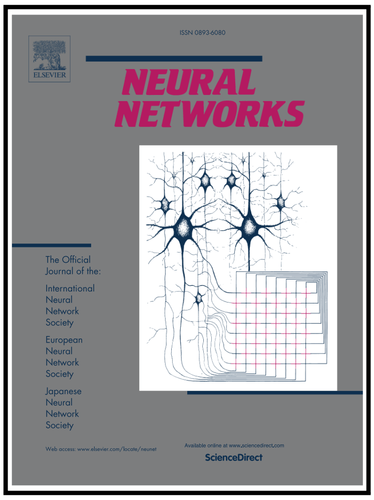

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Structural Knowledge-Guided Complex-Noise Handling for Open Intent Classification

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ABSTRACT

Open intent classification aims to assign known intents to their corresponding classes while identifying unknown intents. Its success largely relies on accurately annotated data. However, real-world data often contains both in-distribution (IND) noise and out-of-distribution (OOD) noise, which degrades model performance. Despite recent progress, open intent classification under such complex-noise scenarios remains underexplored. Moreover, existing noise-handling methods face several limitations: 1) they typically rely on limited class-level information, such as prototypes, for noise detection, overlooking intra-class structural characteristics; 2) they commonly adopt global criteria for sample selection; 3) they tend to ignore the potential value of OOD noise in training. These limitations reduce noise detection accuracy and hinder representation learning. To address these challenges, we propose a structural knowledge-guided **C**omplex-**N**oise Handling method for **O**pen **I**ntent **C**lassification (CNOIC). Specifically, we introduce structural knowledge representation to capture the structural characteristics of each class via granular-balls. Furthermore, we design structure-guided criteria, derived from granular-ball attributes, to distinguish clean samples, IND noise, and OOD noise. Additionally, we present an open space-aware representation learning strategy that uses OOD noise prototypes as placeholders for unknowns, effectively expanding the unknown space. Experiments on three datasets validate the effectiveness of our proposed approach. The source code of CNOIC is available at: https://github.com/Liyanhuaa/CNOIC.

1. Introduction

Traditional intent classification methods are limited to classifying only the predefined known classes encountered during training. In real-world applications, user inputs may contain intents that are not part of the predefined set of known classes. Therefore, it is essential for models to detect such unknown intents. Open Intent Classification (OIC) [31, 49] addresses this need by requiring models to classify the predefined known classes while simultaneously recognizing unknowns. Recent advancements have focused on two primary directions: 1) directly training a $(K+1)$-classifier by adding an extra class for unknown intents [5, 56], and 2) employing a K-classifier combined with an outlier detection mechanism to detect unknowns [21, 50].

Although existing open intent classification methods have achieved satisfactory progress, they rely heavily on large-scale datasets with high-quality annotations. However, in practical applications, datasets are often noisy, as creating such elaborately annotated datasets is both time-consuming and labor-intensive. As illustrated in Fig. 1, training data typically contains three types of samples: clean samples, in-distribution (IND) noise where predefined known-class intents are mislabeled as other known classes, and out-of-distribution (OOD) noise where external intents are incorrectly labeled as known classes. To illustrate these sample types, consider a financial scenario with known classes such as loan and transfer. A clean sample is correctly labeled (e.g., "I want to apply for a loan" labeled as loan), an IND noise sample is mislabeled within known classes (e.g., "Transfer money to my account" labeled as loan), and an OOD noise sample comes from out-of-distribution but is incorrectly assigned to a known class (e.g., "What is the weather today?" labeled as transfer). Furthermore, OOD noise may originate from other datasets (Far-OOD) or from semantically related but unseen intents within the same dataset (Near-OOD). Importantly, the true labels of OOD noise are distinct from the unknown classes encountered during testing. In most real-world scenarios, IND noise and OOD noise often coexist [53], which we refer to as ***complex-noise***. This complex-noise poses a significant challenge to open intent classification, as it may cause severe overfitting and degrade the generalization performance

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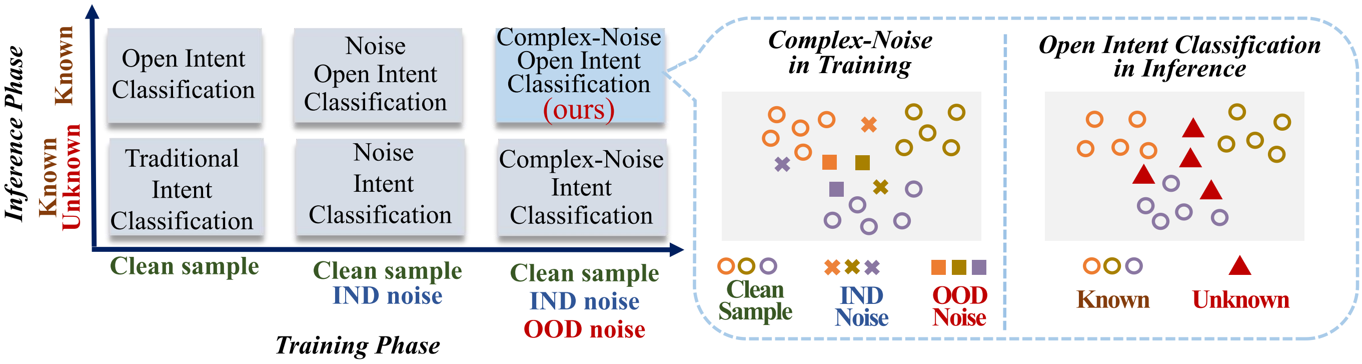


Figure 1: The setting of open intent classification under complex-noise. The training dataset includes clean samples, IND noise, and OOD noise, where different colors indicate their assigned labels. During inference, the model is required to classify known intents and identify unknown ones.

of model [48]. Previous research on noise intent classification has mainly concentrated on closed-world settings [1, 4], which focus on enhancing known intents classification under IND noise. Recently, a study has started to explore open intent classification under pseudo-labeling noise [13], yet it does not explicitly model the coexistence of both IND and OOD noise. The challenge of handling complex-noise during training and recognizing both known and unknown intents during testing remains underexplored.

In addition, other existing noise-handling methods exhibit several common limitations. First, class prototypes are commonly used as class-level knowledge in noisy label learning due to their robustness to noise [55]. However, these prototypes only capture positional knowledge, neglecting other structural characteristics of a class, making distance-based noise identification challenging. Second, sample selection methods are widely employed to distinguish clean samples from noise. Existing approaches often rely on uniform selection criteria, such as fixed thresholds of distance [55] or probability [45]. These coarse-grained global criteria do not account for inter-class differences, making them difficult to fine-tune and poorly suited for real-world data. Third, current noisy label learning methods generally regard OOD noise as harmful and discard it during training [33]. However, OOD noise can serve as a valuable resource in open intent classification by acting as placeholders of unknown classes in representation learning. Without incorporating OOD noise, models trained solely on known classes tend to saturate the representation space with known features, leaving no room for unknown classes, resulting in their misclassification as known during testing.

Motivated by the lack of research on open intent classification under complex-noise and the limitations of existing noise-handling methods, we propose CNOIC, a structural knowledge-guided **C**omplex-**N**oise Handling method for **O**pen **I**ntent **C**lassification (CNOIC). First, we employ granular-ball clustering [35] to encode all representations with multiple granular-balls, which has been proven in our previous work [16] to faithfully capture the true class distributions. By leveraging the multi-dimensional attributes of these granular-balls, we capture both positional and structural knowledge of known classes, enabling more comprehensive characterization of their underlying structure. Next, we propose a structural knowledge-guided noise recognition module. This module leverages the acquired granular-ball-based structural knowledge to establish granular-ball-specific criteria for distinguishing clean samples, IND noise, and OOD noise, thereby reducing the risk of errors associated with applying a single threshold to all samples. Furthermore, we propose an open space-aware representation learning module, which treats the prototypes of OOD noise as placeholders for unknown classes in representation learning. This strategy enables the learned representations to contract the known space and expand the unknown space, which helps improve open intent classification. Finally, during inference, we employ multiple granular-ball decision boundaries to perform open classification. Our main contributions are summarized as follows:

- We propose CNOIC, a unified framework for open intent classification under complex noise. The framework leverages granular-ball clustering to derive structural knowledge, which is used to guide noise recognition and representation learning.

- We design a structural knowledge-guided noise recognition strategy that leverages structural knowledge derived from granular-ball attributes to distinguish clean samples, IND noise, and OOD noise, overcoming the limitations of global thresholding.

- We design an open space-aware representation learning strategy that uses prototypes of OOD noise as proxies for unknown classes, promoting balanced separation between known and unknown spaces and enhancing the detection of novel classes.

2. Related Work

This section provides a detailed review of existing methods related to our task, including open intent classification, learning with noisy labels, and granular-ball computing.

2.1. Open Intent Classification

Open intent classification aims to classify intents from known classes while reliably identifying those that belong to unknown classes during testing. Recently, there has been growing attention to open intent classification [22, 57]. They can be broadly classified into two categories. One approach directly trains a classifier with $(K+1)$-classes, where the additional class denotes unknowns [56]. The other focuses on outlier detection, which is further divided into probability-based or distance-based methods that use predefined thresholds to identify unknowns [15, 28], and boundary-based methods [21, 50], which aim to provide a clear boundary between the known classes and unknowns. Our prior work, MOGB [16], introduced granular-balls to construct multi-granularity decision boundaries for open intent classification.

Despite these advances, most existing methods assume that training data is clean and fully reliable. However, this assumption is often violated in real-world applications, where noisy labels are inevitable. To the best of our knowledge, few studies have considered open intent classification under noisy conditions (OSP-class) [13], which addresses the problem in a weakly supervised setting through pseudo-labeling and noise-robust training. Nevertheless, it does not explicitly model different types of noise or their coexistence. Meanwhile, several studies, such as NGC [33] and CONC [53], have explored open set recognition under complex noise involving both IND and OOD noise. However, these methods mainly rely on graph-based representations developed for image or graph domains, making them less suitable for modeling fine-grained semantic information in textual data. In addition, they primarily focus on filtering unreliable samples and adopt coarse-grained noise modeling strategies, where different types of noise are not explicitly distinguished. Moreover, they do not leverage OOD data as useful signals for representation learning, and thus lack the ability to effectively support the expansion of the open space.

Therefore, open intent classification under complex-noise, where IND and OOD noise coexist in training data while unknown intents appear during testing, remains insufficiently studied. To address this gap, our work focuses on robust open intent classification in the presence of both IND and OOD noise.

2.2. Learning with Noisy Labels

In real-world applications, data annotations are often noisy, making learning from noisy labels a critical challenge. Existing noise-handling approaches mainly focus on computer vision tasks and can be broadly categorized into four types: 1) transition matrix-based methods [20, 39], which model label corruption via estimated noise transitions; 2) label correction methods [40], which relabel noisy samples based on model predictions; 3) loss correction methods [12], which design noise-tolerant loss functions such as Generalized Cross Entropy; and 4) sample selection methods [14, 19], which filter or reweight clean samples using loss statistics or consistency across models.

Despite these efforts, most existing methods, such as Jo-SRC [45] and PNP [29], are designed for noisy label learning under closed-set settings. Recent studies have attempted to improve noise recognition by introducing clustering or multi-prototype representations (e.g., LNL-MPM) [55]. However, these methods typically rely on a predefined number of clusters, which limits their ability to capture the intrinsic structure of data. Moreover, they mainly utilize class-level prototypes for noise identification, while overlooking richer structural characteristics such as class dispersion and boundary variation. In addition, many existing methods rely on uniform sample selection criteria, such as fixed thresholds based on distance or prediction probability [45, 55]. These global criteria ignore local structural variations in the data and fail to adapt to heterogeneous class distributions. Furthermore, existing approaches generally treat OOD noise samples as harmful noise and discard them during training, without exploiting their potential as useful signals for representation learning [33]. As a result, they fail to effectively model the open space and are not well-suited for open intent classification under complex-noise.

Overall, existing noisy label learning methods do not explicitly model the structural characteristics of classes and thus cannot effectively leverage structural information for noise identification. Moreover, OOD samples are typically discarded rather than being exploited in representation learning.

Table 1
Comparison of different methods on open classification, noise handling, and structural modeling capabilities.

Method	Open Classification	IND Noise	OOD Noise	Multi-Prototype Representation	Adaptive Cluster Number	Structural Information	Structure-guided Noise Identification	Beneficial OOD for Open Space
PNP [29]	×	✓	✓	×	×	×	×	×
Jo-SRC [45]	×	✓	✓	×	×	×	×	×
OSP-Class [13]	×	✓	✓	×	×	×	×	×
LNL-MPM [55]	×	✓	✓	✓	×	×	×	×
GBRAIN [32]	×	✓	×	✓	✓	✓	✓	×
GBS [38]	×	✓	×	✓	✓	✓	✓	×
CONC [53]	✓	✓	✓	×	×	✓	×	×
NGC [33]	✓	✓	✓	×	×	×	×	×
CNOIC (Ours)	✓	✓	✓	✓	✓	✓	✓	✓

2.3. Granular-Ball Computing

Granular-ball computing is a robust, adaptive, and interpretable paradigm for multi-granularity representation and computation [35]. It hierarchically constructs data representations, progressing from coarse-grained to fine-grained structures. Each granular-ball encapsulates a group of samples and can be characterized by multiple structural attributes, including its centroid, radius, purity, and size. These attributes enable granular-balls to characterize complex data distributions from multiple perspectives. Granular-ball computing has been widely applied to a variety of learning tasks [44], including granular-ball clustering [41], classification [7, 8, 24], rough set and fuzzy rough set modeling [17, 37, 54], neural network enhancement [27], feature selection [2, 52], graph representation [34], and concept-cognitive learning [9].

Recent studies have shown that granular-ball representations are beneficial for robust classification. For example, M.A. Ganaie et al. [8] proposed the Granular Ball K-Class Twin Support Vector Classifier (GB-TWKSVC), which integrates granular-ball representations with twin support vector machines to improve classification performance and computational efficiency by enhancing spatial locality. In addition, M.A. Ganaie and Vrushank Ahire [7] proposed the Granular Ball Twin Support Vector Machine with Universum Data (GBU-TSVM), which incorporates auxiliary non-target samples to refine decision boundaries and improve robustness and generalization. Tanveer et al. [30] proposed the Granular-Ball Least Square Twin Support Vector Machine (GBLSTSVM), which replaces sample-level inputs with granular-ball representations to enhance the robustness of least square twin support vector machines to noise and outliers. Quadir et al. [25] further proposed the Enhanced Feature-Based Granular-Ball Twin Support Vector Machine (EF-GBTSVM), which enriches granular-ball representations by combining original and randomized hidden features for more discriminative classification. These works demonstrate that granular-ball representations are effective for improving robustness in classification tasks. In addition to such robustness-oriented approaches, recent studies have also explicitly applied granular-ball-based structural information to noisy label learning [42, 51]. For example, GBRAIN [32] groups semantically similar samples together using granular-ball clustering and corrects noisy labels via majority voting, while GBS [38] utilizes granular-ball structures to select boundary samples and mitigate noise and class imbalance. These methods highlight the potential of granular-ball representations for noise handling.

However, existing granular-ball-based methods are primarily designed for closed-set settings and mainly focus on IND noise, without considering OOD noise or unknown classes. In addition, their noise identification strategies typically rely on limited structural attributes, such as centroids or radii, without fully exploiting richer multi-granularity structural information.

In summary, existing methods face significant challenges in handling open intent classification under complex-noise, where both IND and OOD noise coexist during training and unknown intents appear during testing. To address this challenge, we propose a novel framework for open intent classification under complex-noise. As summarized in Table 1, our method differs fundamentally from existing approaches. Specifically, it supports open classification while jointly modeling both IND and OOD noise. Moreover, it leverages granular-ball clustering to automatically discover intra-class structures without requiring a predefined number of clusters. Based on the clustering results, our method captures rich structural information of the data and characterizes the underlying data distribution in a structure-aware manner. In addition, it utilizes structural knowledge derived from granular-ball attributes to guide noise identification, enabling more reliable and fine-grained discrimination between clean samples and different types of noise. Importantly, it treats OOD noise samples not as harmful noise but as beneficial signals for representation learning, thereby facilitating effective modeling and expansion of the open space.

3. Preliminaries

In this section, we introduce the fundamental concepts of granular-ball and present the formal problem setting of open intent classification under complex-noise.

3.1. Granular-Ball

Following previous works [38], we define the fundamental concepts of the granular-ball as follows:

Definition 1. Given a set of representations $Z = \{(z_i, y_i) \mid i = 1, 2, \dots, N\}$, where z_i denotes the feature vector and y_i represents its corresponding class label, a set of granular-balls $G = \{gb_1, gb_2, \dots, gb_m\}$ is constructed on Z. The size n_j of granular-ball gb_j is defined as the total number of samples it encompasses. The center o_j of gb_j is calculated as the mean of all feature vectors it contains, and its radius r_j is defined as the average Euclidean distance from each sample to the center. The o_j and r_j of gb_j are defined as follows:

$$o_j = \frac{1}{n_j}\sum_{i=1}^{n_j} z_i, \quad r_j = \frac{1}{n_j}\sum_{i=1}^{n_j} ||z_i - o_j||, \tag{1}$$

where $||z_i - o_j||$ represents the Euclidean distance between z_i and o_j.

Definition 2. Given the j-th granular-ball $gb_j = \{(z_i, y_i) \mid i = 1, 2, \dots, n_j\}$, let L_j denote the set of class labels associated with samples contained in gb_j, i.e., $L_j = \{y_i \mid (z_i, y_i) \in gb_j\}$. The label of gb_j, denoted as l_j, is defined as the class label with the highest frequency among the samples within the ball:

$$l_j = \arg\max_{l_t \in L_j} \left| \{(z_i, y_i) \in gb_j \mid y_i = l_t\} \right|. \tag{2}$$

The purity of a granular-ball indicates the proportion of samples whose label is equal to l_j, and is defined as follows:

$$p_j = \frac{|\{(z_i, y_i) \in gb_j \mid y_i = l_j\}|}{|gb_j|}, \tag{3}$$

where $|\cdot|$ denotes the cardinality of a set.

3.2. Problem Statement

Let $S = \{(\mu_i, y_i)\}_{i=1}^{N}$ denote the training dataset, where each pair (μ_i, y_i) consists of an input μ_i and its corresponding assigned label $y_i \in I_{\text{Known}}$, with $I_{\text{Known}} = \{1, \cdots, K\}$ representing the set of known classes observed during training. All unknown classes are treated as the $(K+1)$-th class, denoted as $I_{\text{Unknown}} = \{(K+1)\}$. Note that unknown classes only appeared in the testing phase, and thus their feature representations cannot be learned during training. The training dataset contains three types of samples: clean samples, IND noise, and OOD noise. Let y_i^* denote the ground-truth label of sample μ_i. These sample types are defined as follows:

- **Clean samples**: the assigned label y_i matches the ground-truth label y_i^*, i.e., $y_i = y_i^*$ and $y_i^* \in I_{\text{Known}}$.
- **IND noise samples**: the sample belongs to a known class but is mislabeled as another known class, i.e., $y_i \neq y_i^*$ and $y_i^* \in I_{\text{Known}}$.
- **OOD noise samples**: the sample originates from an out-of-distribution class that belongs neither to the known classes nor to the unknown classes, and is incorrectly labeled as a known class, i.e., $y_i \neq y_i^*$ and $y_i^* \notin I_{\text{Known}} \cup I_{\text{Unknown}}$. In our work, we consider two types of OOD noise: Far-OOD noise from external datasets and Near-OOD noise from the same dataset.

In the testing stage, the test dataset consists of samples from both known and unknown classes. The goal of CNOIC is to learn robust representations from complex-noise data, accurately classify known intents into their corresponding classes I_{Known}, and recognize open samples as I_{Unknown}.

4. Methodology

This section presents the overall methodology of CNOIC. We first describe the framework architecture and then introduce its three core modules in detail, including structural knowledge representation, complex-noise recognition, and open space-aware representation learning. Finally, we present the inference mechanism and the overall algorithmic procedure of the proposed method.

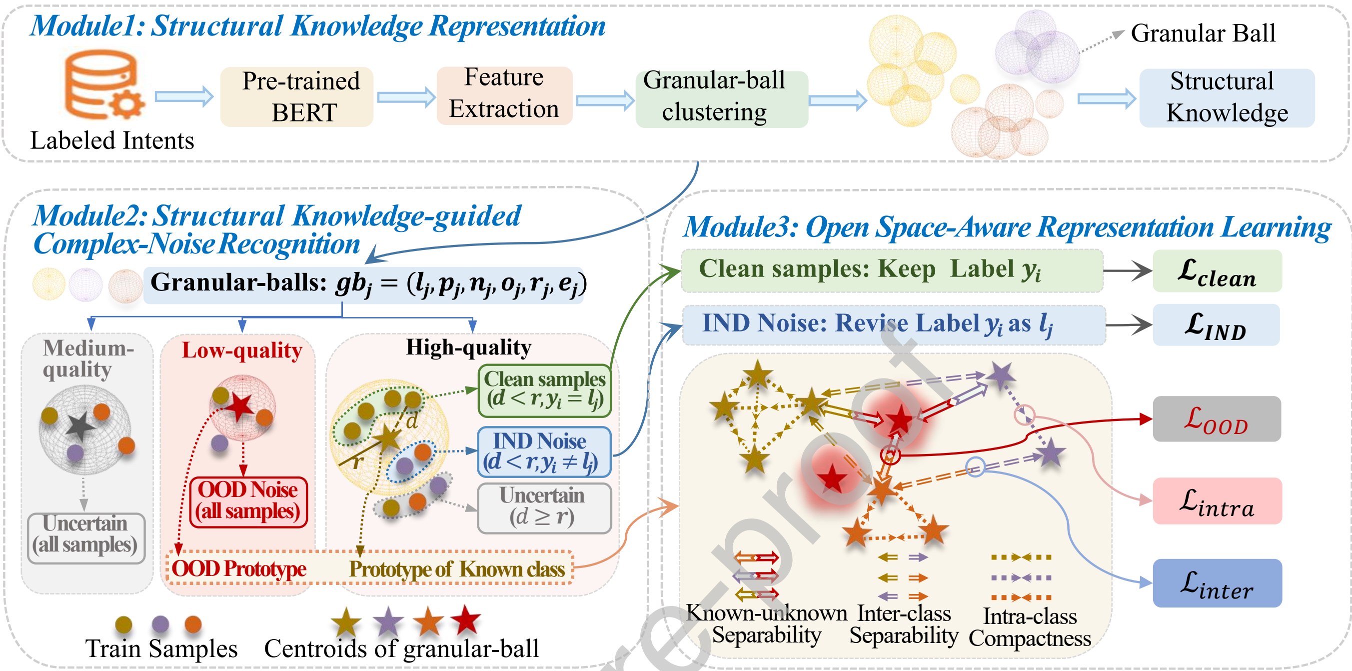


Figure 2: The overview of CNOIC. Module 1 extracts structural knowledge from labeled samples through granular-ball clustering; Module 2 identifies clean samples, IND noise, and OOD noise based on structural knowledge; Module 3 improves feature robustness through open space-aware representation learning.

4.1. Overview of the Proposed Method

The proposed CNOIC method consists of three modules, as illustrated in Fig. 2. It begins with a structural knowledge representation module to capture rich characteristics of known classes. Based on these representations, a structural knowledge-guided noise recognition module is constructed to distinguish clean samples, IND noise, and OOD noise. Furthermore, an open space-aware learning module is employed to learn robust representations. After representation learning, multi-granularity decision boundaries [16] derived from structural knowledge are utilized to classify known intents and detect unknown intents. Each module is elaborated as follows.

4.2. Structural Knowledge Representation

Noise recognition methods that rely solely on multi-prototypes of classes often fail to leverage richer structural information within classes, which hinders the accurate identification of noisy samples [55]. To overcome this limitation, we aim to acquire more comprehensive and noise-robust representations of known classes to support noise recognition. Following granular-ball computing [36] and granular-ball representation in our previous work [16], we leverage granular-ball clustering to generate varying-size granular-balls that reflect the true structure of the dataset. Then, each known class is represented by multiple granular-balls of varying sizes. This comprehensive information derived from the granular-balls forms the structural knowledge of each known class.

Initially, all feature representations $Z = \{(z_i, y_i) \mid i = 1, 2, \ldots, N\}$ of known classes are extracted using a pre-trained BERT model [11]. Subsequently, these representations are input into a granular-ball clustering algorithm, which is an iterative process that continually splits granular-balls into finer-grained ones until a stopping condition is met. The process begins by treating the entire set Z as a single granular-ball. This ball is evaluated based on two criteria: purity p_l and size n_l. If the purity is below p_l and the size exceeds n_l, the ball will be split. Specifically, for a granular-ball gb_j that contains samples belonging to multiple classes, we identify all unique labels within it and subdivide the ball into several new granular-balls, each corresponding to one of these unique labels. Initial centers for these new balls are established by randomly selecting pseudo-centroids from samples of their respective classes. Each sample in gb_j is assigned to the nearest new ball based on Euclidean distance to the pseudo-centroids. After reassignment, the purity and size of each new granular-ball are assessed. Balls meeting the stopping criteria (purity $> p_l$ or size $< n_l$) cease further splitting. Others undergo additional iterations. This recursive process continues until all granular-balls satisfy the stopping conditions.

After granular-ball clustering, a set of granular-balls $G = \{gb_1, gb_2, \ldots, gb_m\}$ is generated, where each granular-ball contains a set of samples $e_j = \{(z_i, y_i) \mid i = 1, \ldots, n_j\}$. For each granular-ball gb_j, we compute its size n_j, assigned label l_j, purity p_j, centroid o_j, and radius r_j, as defined in Definitions 1 and 2. These attributes together describe granular-ball gb_j as:

$$gb_j = (l_j, p_j, n_j, o_j, r_j, e_j). \tag{4}$$

For each known class $c \in I_{\text{Known}}$, the set of granular-balls with label $l_j = c$ constitutes the structural knowledge of that class. The attributes of these granular-balls jointly characterize the positional and distributional properties of class c from multiple perspectives. Compared with single-prototype representations that rely on a single global centroid, such a representation models each class via multiple localized structural units, enabling the approximation of complex and irregular class distributions through several compact local regions, and thereby providing a more faithful characterization of the intrinsic class structure. Based on this structural knowledge, we establish a solid foundation for subsequent complex-noise recognition.

4.3. Structural Knowledge-guided Complex-Noise Recognition

Existing threshold-based noise recognition methods [45, 55] typically apply a uniform global threshold to the entire dataset. However, this global rule fails to capture inter-class structural differences and ignores region-specific characteristics in the feature space. This limitation undermines the accuracy of noise identification and may further impair the effectiveness of subsequent representation learning. To overcome this limitation, we leverage the learned structural knowledge to perform complex-noise recognition. This adaptive criterion operates at the granular-ball level, where the noise types of samples are determined based on the attributes of their respective granular-balls. This ball-wise strategy is more aligned with actual data distributions, enhancing the accuracy of noise detection.

Specifically, our approach is motivated by the observation that different noise types exhibit distinct distribution patterns in the granular-ball representation space. In general, clean samples and IND noise, both originating from known classes, tend to distribute around the reliable and core regions of known classes. In contrast, OOD noise samples, which originate from unknown classes, are more likely to appear in peripheral regions that deviate from known-class distributions [55]. Consequently, clean samples and IND noise are more likely to reside in the core region of high-purity granular-balls that correspond to known classes, while OOD noise tends to fall into low-purity granular-balls that fail to form coherent known-class structures. Based on this observation, we divide all granular-balls into three quality levels according to their purity values: high-quality ($p_j > p_h$), medium-quality ($p_w < p_j \leq p_h$), and low-quality ($p_j \leq p_w$). Different noise identification strategies are then applied to samples contained in granular-balls of different quality levels.

High-quality granular-balls are regarded as effective representations of known classes due to their high label consistency. For each such granular-ball, its centroid o_j indicates one of the structural centers of the corresponding known-class region, and its radius r_j is used as a reference to approximate the core region of that granular-ball. For samples contained in a high-quality granular-ball, those located within the radius are considered to lie in the core region of the ball and are therefore more likely to originate from the corresponding known class. Such samples may either be clean samples or IND noise. Specifically, samples whose labels are consistent with the granular-ball label (i.e., $y_i = l_j$) are identified as clean samples, whereas those with inconsistent labels (i.e., $y_i \neq l_j$) are regarded as IND noise. By contrast, samples located outside the radius lie farther away from the core region of the high-quality granular-ball. Although they are included in a granular-ball associated with a known class, their true types remain ambiguous due to their peripheral positions. To avoid introducing unreliable supervision, such samples are conservatively categorized as uncertain samples. Based on the above criteria, the corresponding decision rule is given as follows:

$$(z_i, y_i) \in \begin{cases} \text{Clean sample,} & \text{if } \|z_i - o_j\| < r_j \text{ and } y_i = l_j, \\ \text{IND noise,} & \text{if } \|z_i - o_j\| < r_j \text{ and } y_i \neq l_j, \\ \text{Uncertain sample,} & \text{if } \|z_i - o_j\| \geq r_j, \end{cases} \tag{5}$$

where $\|z_i - o_j\|$ denotes the Euclidean distance between sample z_i and centroid o_j.

Medium-quality granular-balls exhibit moderate label consistency, making it difficult to confidently determine the true noise type of the samples they include. To prevent erroneous supervision, all samples included in such granular-balls are conservatively labeled as uncertain samples and excluded from subsequent training stages.

Low-quality granular-balls exhibit low label consistency and are assumed to consist of samples unrelated to any known class. Therefore, all included samples in such granular-balls are directly classified as OOD noise.

In summary, let $\mathcal{M}_{\text{clean}}$, $\mathcal{M}_{\text{IND}}$, $\mathcal{M}_{\text{OOD}}$, and $\mathcal{M}_{\text{uncertain}}$ denote the sets of clean samples, IND noise, OOD noise, and uncertain samples. These sets are derived based on the previously defined noise identification rules, as follows:

$$\mathcal{M}_{\text{clean}} = \bigcup_{gb_j \in G} \left\{ (z_i, y_i) \in e_j \,\middle|\, p_j > p_h \wedge \|z_i - o_j\| < r_j \wedge y_i = l_j \right\}, \tag{6}$$

$$\mathcal{M}_{\text{IND}} = \bigcup_{gb_j \in G} \left\{ (z_i, y_i) \in e_j \,\middle|\, p_j > p_h \wedge \|z_i - o_j\| < r_j \wedge y_i \neq l_j \right\}, \tag{7}$$

$$\mathcal{M}_{\text{OOD}} = \bigcup_{gb_j \in G} \left\{ (z_i, y_i) \in e_j \,\middle|\, p_j \leq p_w \right\}, \tag{8}$$

$$\mathcal{M}_{\text{uncertain}} = \bigcup_{gb_j \in G} \left\{ (z_i, y_i) \in e_j \,\middle|\, (p_j > p_h \wedge \|z_i - o_j\| \geq r_j) \vee (p_w < p_j \leq p_h) \right\}. \tag{9}$$

Derived from comprehensive positional and structural knowledge, these ball-wise noise recognition rules enable adaptive and effective identification of diverse noise types. Additionally, we obtain the centroids and labels of high-quality granular-balls to represent the prototypes of known classes, denoted as $\mathcal{N}_k$, and centroids of low-quality granular-balls to represent the prototypes of OOD noise, denoted as $\mathcal{N}_u$. These prototypes are used to enhance subsequent representation learning. $\mathcal{N}_k$ and $\mathcal{N}_u$ are generated as follows:

$$\begin{aligned} \mathcal{N}_k &= \{(o_j, l_j) \mid gb_j \in G,\ p_j > p_h\}, \\ \mathcal{N}_u &= \{o_j \mid gb_j \in G,\ p_j \leq p_w\}. \end{aligned} \tag{10}$$

4.4. Open Space-Aware Representation Learning

After identifying clean samples, IND noise, and OOD noise, we aim to learn balanced representations of both known and unknown intent spaces to facilitate open intent classification. To this end, we introduce OOD prototypes as auxiliary structural anchors to regularize the representation space, encouraging the model to form compact known regions while maintaining clear separation from the surrounding feature space. In this way, the model is guided to reserve sufficient space for potential unknown patterns without explicitly modeling specific OOD samples. In our framework, the open space is not explicitly defined by OOD instances, but is implicitly characterized as the complement of the compact regions formed by known classes. To effectively learn such representations, we design distinct training objectives for different types of samples and prototypes:

Clean samples $(z_i, y_i) \in \mathcal{M}_{\text{clean}}$ are trained using standard cross-entropy loss with their original assigned labels y_i:

$$\mathcal{L}_{\text{clean}} = -\frac{1}{|\mathcal{M}_{\text{clean}}|} \sum_{(z_i, y_i) \in \mathcal{M}_{\text{clean}}} \log p(y_i | z_i). \tag{11}$$

These samples provide standard supervision and reinforce discriminative features for known classes.

IND noise samples $(z_i, y_i) \in \mathcal{M}_{\text{IND}}$ are relabeled as the label l_j of the granular-ball gb_j they belong to. The corrected labels are then used to compute cross-entropy loss:

$$\mathcal{L}_{\text{IND}} = -\frac{1}{|\mathcal{M}_{\text{IND}}|} \sum_{(z_i, y_i) \in \mathcal{M}_{\text{IND}}} \log p(l_j | z_i). \tag{12}$$

This correction mitigates noisy gradient signals and aligns IND samples with their true semantic structure.

Prototypes of OOD noise are introduced as auxiliary structural cues to characterize the surrounding feature space. To facilitate open space awareness, the known class prototypes $(o_k, l_k) \in \mathcal{N}_k$ are encouraged to remain well-separated from the OOD prototypes $o_u \in \mathcal{N}_u$, as formulated below:

$$\mathcal{L}_{\text{OOD}} = \frac{1}{|\mathcal{N}_u| \cdot |\mathcal{N}_k|} \sum_{u=1}^{|\mathcal{N}_u|} \sum_{k=1}^{|\mathcal{N}_k|} \frac{1}{\|o_u - o_k\| + \epsilon}, \tag{13}$$

where ϵ is a small constant (e.g., 10^{-6}) to prevent numerical instability. This constraint enforces a clear margin between compact known regions and the surrounding feature space, thereby facilitating better separation during inference.

To further refine the structure of the known representation space, we introduce two additional prototype-level constraints aimed at enhancing intra-class compactness and inter-class separability. Specifically, prototypes belonging to the same known class are pulled closer together. Let o_i and o_k denote prototypes of known classes in $\mathcal{N}_k$, the intra-class compactness loss is defined as:

$$\mathcal{L}_{\text{intra}} = \frac{1}{|\mathcal{N}_k| \cdot (|\mathcal{N}_k| - 1)} \sum_{i=1}^{|\mathcal{N}_k|} \sum_{\substack{k=1 \\ k \neq i}}^{|\mathcal{N}_k|} \mathbb{I}[l_i = l_k] \cdot \|o_i - o_k\|, \tag{14}$$

where $\mathbb{I}[\cdot]$ is an indicator function returning 1 if the condition holds and 0 otherwise. This encourages tight, coherent clusters within each known class. Conversely, prototypes from different known classes are pushed apart to improve inter-class separability and enhance class discriminability in the learned feature space. The inter-class separation loss is defined as follows:

$$\mathcal{L}_{\text{inter}} = \frac{1}{|\mathcal{N}_k| \cdot (|\mathcal{N}_k| - 1)} \sum_{i=1}^{|\mathcal{N}_k|} \sum_{\substack{k=1 \\ k \neq i}}^{|\mathcal{N}_k|} \mathbb{I}[l_i \neq l_k] \cdot \frac{1}{\|o_i - o_k\| + \epsilon}. \tag{15}$$

The overall training objective integrates all components in a unified manner:

$$\mathcal{L} = \mathcal{L}_{\text{clean}} + \mathcal{L}_{\text{IND}} + \mathcal{L}_{\text{OOD}} + \mathcal{L}_{\text{intra}} + \mathcal{L}_{\text{inter}}. \tag{16}$$

This comprehensive loss formulation integrates sample-level supervision with prototype-level structural constraints, enabling the model to denoise labels, preserve class-wise structure, and remain aware of open space. By explicitly incorporating clean samples, IND noise, and OOD noise into training, the open space-aware representation learning module enhances robustness under complex-noise conditions and establishes a strong foundation for reliable detection of unknown classes during inference.

It is worth noting that, different from existing methods that generate pseudo-OOD samples or fix OOD prototypes at the early stage of training, the proposed framework adopts a dynamic and structure-aware mechanism. In our method, granular-balls are re-constructed at each training epoch based on the updated representations, and both known and OOD-related prototypes are continuously updated according to the evolving feature space. As a result, the OOD prototypes are not tied to specific samples or fixed distributions, but instead serve as adaptive structural cues that reflect the current geometry of the representation space. This dynamic update mechanism effectively alleviates the risk of overfitting to particular OOD samples and enables the model to learn a more generalizable separation between compact known regions and the surrounding open space. Therefore, compared with methods that rely on static pseudo-OOD generation, the proposed approach provides a more flexible and robust way to characterize open space, which is particularly beneficial under complex noise conditions.

4.5. Open Intent Classifier in Inference

The MOGB method [16] has demonstrated the effectiveness of multi-granularity decision boundaries in open intent classification. The multi-granularity decision boundary enables better approximation of the diverse spatial structures of known classes and rejects unknown samples that fall outside these regions. In this work, we employ this multi-granularity decision boundary strategy.

Specifically, during inference, we construct adaptive decision boundaries for each known class by leveraging their high-purity and sufficiently large granular-balls (satisfying $p_j > p_t$ and $n_j > n_t$) formed on the learned representations. For a given class $c \in I_{\text{Known}} = \{1, \ldots, K\}$, we collect all associated granular-balls, each characterized by a centroid o_s^c and a radius r_s^c, where $s = 1, \ldots, n_c$, n_c denotes the number of granular-balls of class c. The collection of tuples $\{o_s^c, r_s^c\}_{c,s=1}^{K,n_c}$ serves as the multi-granularity boundaries of known classes.

In inference, for a query representation t_i, we compute its distances to all known class centroids $\{o_s^c\}$. If t_i lies outside the radii of all granular-balls, meaning it does not fall within any known decision region, it is classified as unknown (K+1). Otherwise, it is assigned to the class corresponding to the nearest granular-ball in terms of distance. Formally, the decision rule is defined as:

$$\hat{y}_i = \begin{cases} K+1, \text{if } \|t_i - o_s^c\| > r_s^c,\ \forall c \in I_{\text{Known}}, \\ \underset{c \in I_{\text{Known}}, \|t_i - o_s^c\| \leq r_s^c}{\arg\min} \|t_i - o_s^c\|, \text{Otherwise.} \end{cases} \tag{17}$$

By relying on class-specific granular-balls to describe regions in the feature space, it effectively captures the underlying structure of known classes. This improves the model's capability to classify known intents and reject unknowns, making it well-suited for real-world open scenarios with noisy data.

4.6. Algorithm Design

Our proposed CNOIC method is illustrated in Algorithms 1. First, input samples are encoded using a pre-trained BERT model to obtain feature representations. In Module 1, these features are then clustered into granular-balls via iterative splitting based on purity and size thresholds, producing structural knowledge of known classes. Based on the constructed knowledge, Module 2 identifies the noise types of training samples, including clean samples, IND noise, OOD noise, and uncertain samples. The classification rule is based on the knowledge derived from the attributes of each granular-ball, allowing for adaptive and structure-aware noise recognition. Module 3 designs distinct loss functions for each type of noise. Clean samples and IND noise are optimized using cross-entropy losses, while contrastive and structural constraints are introduced to enhance the separation between known and unknown prototypes and refine class discriminability. The total loss is used to update the model parameters. This training process is repeated across multiple rounds: updated representations are re-encoded, re-clustered, and re-evaluated through Modules 1 to 3 until convergence. The final optimized model parameters are then used for inference. In the inference stage, we conduct open intent classification by multi-granularity decision boundaries. A test sample is labeled as unknown if it falls outside all boundaries of known classes; otherwise, it is assigned to the nearest known class based on distance.

5. Experiments

In this section, we conduct comprehensive experiments to evaluate the proposed CNOIC method. We first introduce the experimental setup, followed by extensive results under various noise conditions and openness settings. Finally, ablation studies and parameter analyses are conducted to further validate the effectiveness and robustness of the proposed method.

5.1. Experimental Setup

In this subsection, we present the experimental setup for evaluating CNOIC, including the datasets, open and noisy settings, baseline methods, evaluation metrics, and implementation details.

5.1.1. Datasets

We adopt three widely used benchmark datasets in the open intent classification literature to comprehensively evaluate the proposed CNOIC. These datasets cover three representative application domains, namely financial

Algorithm 1: Training of CNOIC.

Input: Feature set $Z = \{(z_i, y_i)\}_{i=1}^{N}$.

Module 1: Structural Knowledge Representation

Initialize queue $Q \leftarrow \{Z\}$ and result set $G \leftarrow \emptyset$;
while *Q is not empty* **do**
  Pop a granular-ball gb_j from Q and compute p_j, n_j;
  if $p_j > p_l$ ***or*** $n_j < n_l$ **then**
    Add gb_j to G;
  else
    Identify unique labels in gb_j;
    Initialize pseudo-centroids for new granular-balls;
    for $(z_i, y_i) \in gb_j$ **do**
      Assign (z_i, y_i) to nearest new granular-ball;
    Add all new granular-balls to Q.
Get $G =_{\{gb_1, gb_2, \ldots, gb_m\}}$ as structural knowledge.

Module 2: Structural Knowledge-guided Complex-Noise Recognition

for gb_j *in* $G = \{gb_1, gb_2, \ldots, gb_m\}$ **do**
  if $p_j > p_h$ **then**
    Classify all samples (z_i, y_i) in gb_j as clean samples, IND noise, uncertain sample by Equation 5;
  if $p_w < p_j \leq p_h$ **then**
    Classify (z_i, y_i) in gb_j as uncertain samples;
  if $p_j \leq p_w$ **then**
    Classify (z_i, y_i) in gb_j as OOD noise;
Collect prototypes $\mathcal{N}_k$ and $\mathcal{N}_u$ by Equation 10.

Module 3: Open Space-Aware Representation Learning

Compute $\mathcal{L}_{\text{clean}}$ for clean samples in $\mathcal{M}_{\text{clean}}$ by Equation 11;
Compute $\mathcal{L}_{\text{IND}}$ for IND noise in $\mathcal{M}_{\text{IND}}$ by Equation 12;
Compute $\mathcal{L}_{\text{OOD}}$ by Equation 13;
Compute $\mathcal{L}_{\text{intra}}$ by Equation 14;
Compute $\mathcal{L}_{\text{inter}}$ by Equation 15;
Overall Loss: $\mathcal{L} = \mathcal{L}_{\text{clean}} + \mathcal{L}_{\text{IND}} + \mathcal{L}_{\text{OOD}} + \mathcal{L}_{\text{intra}} + \mathcal{L}_{\text{inter}}$.

Output: Trained Model.

Table 2
Statistics of datasets.

Dataset	*#Classes*	*#Training*	*#Validation*	*#Test*	*Vocabulary Size*	*Mean Length*
BANKING	77	9,003	1,000	3,080	5,028	11.91
SNIPS	7	13,084	700	700	11,971	9.05
StackOverflow	20	12,000	2,000	6,000	17,182	9.18

services, virtual assistants, and programming question answering. They also differ substantially in the number of intent classes, vocabulary size, semantic characteristics, and dataset scale, providing diverse evaluation scenarios for assessing the robustness and generalization ability of the proposed method. A summary of the datasets is provided below, with detailed statistics shown in Table 2. Here, *Vocabulary Size* denotes the number of unique tokens in the dataset, and *Mean Length* refers to the average number of tokens per utterance.

- **BANKING** [3]: Includes 13,083 online banking query texts across 77 classes, designed for financial services.

- **SNIPS** [6]: Contains 14,484 utterances for a voice assistant, covering 7 classes for everyday tasks.
- **StackOverflow** [43]: A subset of 20,000 programming question titles with 20 classes.

5.1.2. Open & Noise Setting

Some existing open intent classification methods simulate unknown classes by treating the last class in the dataset as unknown and using the remaining classes as known [53]. However, this setting does not align well with real-world applications. To better reflect realistic open-world scenarios, we adopt a more general open setting by randomly selecting a fixed ratio of classes as known classes. The remaining classes are evenly split into two groups: one is used as the OOD classes to construct OOD noise during training, while the other is held out as unknown classes for evaluation during testing.

The training set includes a mixture of clean samples, IND noise, and OOD noise. All samples in the training set, including noisy ones, are assigned labels from the known class set. Specifically, in our default setting, IND noise follows a uniform symmetric noise model, where noisy samples are randomly selected from known-class data and their labels are uniformly flipped to other known classes. For OOD noise, samples are drawn from out-of-distribution data and randomly assigned labels from the known class set. Therefore, both IND and OOD noise are injected under a uniform distribution. To simulate realistic OOD noise more effectively, we introduce two types of OOD noise:

- **Near-OOD noise** is constructed by sampling from the OOD classes of the same dataset and randomly labeling them as known classes.
- **Far-OOD noise** is generated by sampling from other different datasets and assigning these samples with random known class labels.

To explore the effect of Far-OOD noise, we simulate it by sampling from a different dataset. Specifically, we use SNIPS as the noise source when training on BANKING and StackOverflow, and StackOverflow as the noise source when training on SNIPS. The test set contains samples from both the known and unknown classes, where the unknown classes are never seen during training. To evaluate model robustness under diverse open-world and noisy conditions, we consider different known class ratios (25%, 50%) and noise ratios (20%, 40%), with each noise setting comprising an equal proportion of IND and OOD noise and all noise is uniformly distributed across classes in the default setting. We also conduct comparative evaluations under both Near-OOD and Far-OOD settings to comprehensively assess the performance of our proposed method.

5.1.3. Baseline Methods

To evaluate the performance of CNOIC, we compare it with two categories of baseline methods. The first type of baselines is open intent classification methods, which focus on identifying unknown samples and classifying known ones, including **DeepUnk** [18], **K+1-way** [47], **OLT** [58], **DETER** [46], and the recent **Ellipsoid** [59] method, which models class boundaries using ellipsoidal decision regions to better capture complex data distributions and improve open-set recognition performance. The second category of baselines consists of noisy-label learning methods, including **GBRAIN** [32]. In addition, we include a KNN-based label noise analysis method, denoted as **LN-KNN** [26], for comparison. Since these methods are not originally designed for unknown intent detection, we equip them with an adaptive boundary to enable unknown intent identification [49].

5.1.4. Evaluation Metrics

We employ four evaluation metrics to assess performance. F1-All denotes the macro F1-score computed over both known and unknown classes, while ACC refers to the overall classification accuracy. To provide a more fine-grained analysis, we additionally report the macro F1-score for known classes (F1-Known or F1-K) and the F1-score for unknown classes (F1-Unknown or F1-U).

5.1.5. Implementation Details

For representation learning, we adopt a pre-trained BERT model [11] with all parameters frozen except for the final classification layer. The model is trained with a batch size of 128 and a learning rate of 2×10^{-5}. The training is conducted for 100 epochs, and an early stopping strategy is applied, where training is terminated if the validation performance does not improve for 10 consecutive epochs. In the granular-ball clustering stage, we follow standard practice by setting the purity threshold to $p_l = 0.9$. The size threshold n_l is selected based on the average number of

Table 3
Results of open intent classification across BANKING, SNIPS, and StackOverflow with different known class ratios (25%, 50%) and noise ratios (20%, 40%) under IND noise and **Near-OOD Noise**. The best results are highlighted in bold, and the second-best results are underlined.

	Datasets	BANKING				SNIPS				StackOverflow			
	Noise%	20%		40%		20%		40%		20%		40%	
Known%	Methods	ACC	F1-All	ACC	F1-All	ACC	F1-All	ACC	F1-All	ACC	F1-All	ACC	F1-All
25%	GBRAIN [32]	27.61	33.63	24.48	30.58	39.85	37.53	36.26	33.84	53.87	60.52	38.24	49.37
	LN-KNN [26]	57.11	27.41	57.63	27.71	58.77	54.56	67.74	63.62	41.11	27.93	42.64	28.63
	DeepUnk [18]	51.00	58.07	46.99	52.75	54.14	55.15	49.07	48.37	46.99	50.24	43.93	47.97
	K+1-way [47]	79.24	75.73	70.01	69.77	80.42	81.23	75.16	77.15	65.05	68.61	58.15	63.70
	OLT [58]	39.22	46.91	38.28	57.37	52.05	52.20	51.63	52.97	39.18	49.47	37.71	46.48
	DETER [46]	45.47	56.29	40.44	45.29	78.51	80.55	78.79	82.78	52.28	57.11	57.91	61.73
	Ellipsoid [59]	62.46	56.42	57.66	51.54	71.77	64.75	70.34	63.69	72.82	56.58	73.88	58.67
	CNOIC (Ours)	**81.67**	**80.25**	**83.43**	**77.46**	**88.34**	**89.21**	**86.30**	**86.36**	**78.19**	**77.62**	**82.27**	**80.68**
50%	GBRAIN [32]	43.61	44.62	40.92	42.22	72.56	65.61	64.55	58.78	53.85	65.11	55.19	62.80
	LN-KNN [26]	39.28	26.29	39.34	25.70	44.16	44.36	41.25	39.16	29.88	20.81	30.44	20.96
	DeepUnk [18]	67.73	71.07	62.67	65.77	85.93	85.62	80.48	80.52	58.68	62.29	55.55	59.21
	K+1-way [47]	76.60	74.63	72.08	75.04	82.34	83.20	80.06	73.57	73.48	75.71	65.50	69.08
	OLT [58]	62.62	76.11	66.55	75.02	75.26	71.10	77.13	71.91	60.55	69.85	59.64	69.71
	DETER [46]	61.72	69.89	62.16	69.47	81.07	83.57	79.29	82.27	65.74	69.11	67.03	71.49
	Ellipsoid [59]	52.84	46.71	51.60	45.63	43.06	40.07	40.22	38.91	64.98	51.49	65.46	52.27
	CNOIC (Ours)	**79.52**	**82.40**	**74.29**	**75.17**	**92.62**	**92.83**	**84.79**	**85.68**	**82.43**	**84.26**	**82.23**	**83.75**

samples per class in each dataset to control the granularity of clustering: $n_l = 20$ for SNIPS, $n_l = 10$ for StackOverflow, and $n_l = 8$ for BANKING. For structural knowledge-guided noise recognition, we use a unified high-quality purity threshold of $p_h = 0.8$ across all datasets to identify high-quality granular-balls. The low-quality purity threshold is dataset-specific, with $p_w = 0.3$ for SNIPS, $p_w = 0.5$ for StackOverflow, and $p_w = 0.4$ for BANKING. During inference, decision boundaries for each known class are constructed using granular-balls that satisfy the conditions $p_j > p_t$ and $n_j > n_t$. The purity threshold p_t is set to 0.6 across all datasets. The size threshold n_t is dataset-specific, set to 4 for BANKING, 9 for StackOverflow, and 80 for SNIPS.

5.2. Experimental Results

In this subsection, we evaluate the proposed CNOIC method under a variety of challenging settings. Specifically, we first report the overall classification performance under Near-OOD and Far-OOD noise. We then examine the robustness of the method under different noise distributions and real-world noisy datasets. Furthermore, we provide fine-grained analysis on known and unknown class recognition, followed by statistical significance tests and visualization results, to further validate the effectiveness of the proposed method.

5.2.1. Performance under IND Noise and Near-OOD Noise

Table 3 presents the F1-All and ACC of CNOIC and several baseline methods under a complex-noise setting that includes both IND noise and Near-OOD noise. The evaluations are conducted on three benchmark datasets across two known class ratios (25% and 50%) and two noise levels (20% and 40%), with each noise level containing equal proportions of IND and Near-OOD noise.

As shown in Table 3, CNOIC consistently achieves the best performance across nearly all settings, highlighting its strong robustness and generalization capability in the presence of complex noise. Compared with the second-best methods, CNOIC achieves consistent improvements across datasets. For example, on the BANKING dataset with 25% known classes and 20% noise, CNOIC improves F1-All from 75.73% to 80.25% (+4.52%). On the SNIPS dataset under the same setting, the improvement is from 81.23% to 89.21% (+7.98%). Similarly, on the StackOverflow dataset with 40% noise, CNOIC improves F1-All from 63.70% to 80.68% (+16.98%). These results further support the effectiveness of the proposed method.

In addition, we observe an interesting pattern regarding the influence of the known class ratio under different noise levels. When the noise ratio is relatively low (20%), the model tends to achieve better performance with a higher known

Table 4
Results of open intent classification across BANKING, SNIPS, and StackOverflow with different known class ratios (25%, 50%) and noise ratios (20%, 40%) under IND noise and **Far-OOD Noise**.

	Datasets	BANKING				SNIPS				StackOverflow			
	Noise%	20%		40%		20%		40%		20%		40%	
Known%	Methods	ACC	F1-All	ACC	F1-All	ACC	F1-All	ACC	F1-All	ACC	F1-All	ACC	F1-All
25%	GBRAIN [32]	34.65	30.77	31.21	28.53	49.85	43.95	51.41	45.44	44.74	48.97	47.39	52.49
	LN-KNN [26]	57.20	28.32	57.74	28.39	58.22	50.69	59.70	51.37	41.34	28.26	43.02	28.87
	DeepUnk [18]	32.03	36.95	28.52	33.90	59.26	60.78	54.12	56.77	41.10	45.53	36.50	39.39
	K+1-way [47]	69.71	71.75	61.84	63.79	82.97	74.48	62.13	66.41	61.23	67.79	58.27	64.97
	OLT [58]	38.07	56.47	34.61	52.26	50.14	47.35	49.50	47.11	59.24	63.12	59.24	62.48
	DETER [46]	74.82	74.57	71.25	62.98	**92.57**	**93.70**	83.55	**87.24**	75.61	**79.11**	72.80	76.23
	Ellipsoid [59]	59.09	53.47	62.13	56.16	55.24	45.25	64.86	53.91	73.29	57.10	72.28	55.81
	CNOIC (Ours)	**79.99**	**79.03**	**82.63**	**78.37**	80.78	82.26	**86.27**	86.07	**76.03**	76.02	**80.91**	**79.17**
50%	GBRAIN [32]	50.64	43.46	45.94	39.14	79.54	76.30	74.38	67.13	72.71	71.98	67.08	69.17
	LN-KNN [26]	38.46	25.45	38.14	25.34	39.98	37.42	38.98	37.01	28.75	22.77	29.38	22.96
	DeepUnk [18]	66.27	70.29	61.24	63.82	70.15	71.33	64.30	65.73	65.59	70.09	59.65	64.12
	K+1-way [47]	73.67	75.55	69.22	71.79	79.78	71.34	73.00	62.08	74.18	77.55	68.15	74.65
	OLT [58]	61.57	75.62	57.56	70.50	72.25	64.31	74.83	66.52	60.66	66.81	60.62	67.44
	DETER [46]	**81.19**	79.02	**79.40**	**78.06**	78.03	81.93	77.17	79.71	78.15	79.25	80.78	82.32
	Ellipsoid [59]	44.75	38.58	45.86	39.58	41.00	40.53	37.84	37.13	65.28	51.49	64.74	50.81
	CNOIC (Ours)	80.31	**83.13**	76.10	77.95	**89.38**	**89.59**	**89.20**	**89.79**	**80.76**	**83.09**	**82.49**	**84.03**

class ratio (50%). In contrast, when the noise ratio increases to 40%, the model performs better under a lower known class ratio (25%). This phenomenon can be attributed to the trade-off between class diversity and noise complexity. Under low-noise conditions, increasing the number of known classes provides richer semantic information and helps learn more discriminative representations. However, under high-noise conditions, a larger number of classes introduces more complex noise patterns and increases the difficulty of distinguishing clean samples from noisy ones. In this case, a smaller number of known classes leads to more stable class structures, which facilitates more robust learning.

5.2.2. Performance under IND noise and Far-OOD Noise

To investigate the impact of different types of OOD noise, we conducted additional experiments wherein the OOD noise is Far-OOD noise, which refers to instances sampled from entirely different datasets. All other experimental settings were kept consistent with those described in Sec. 5.2.1.

The results of these experiments are summarized in Table 4. CNOIC achieves consistently strong performance across different datasets and settings under Far-OOD noise. Different from the Near-OOD setting, the performance remains relatively stable when varying both the noise ratio and the known class ratio. Increasing the noise ratio from 20% to 40% does not lead to consistent performance degradation, and in some cases, even results in slight improvements (e.g., on SNIPS and StackOverflow). Similarly, changing the known class ratio does not cause obvious performance drops. This can be attributed to the nature of Far-OOD noise. Since Far-OOD samples are typically drawn from different domains and are semantically distant from known classes, they are easier to separate in the representation space and thus have a relatively limited impact on model performance.

5.2.3. Performance under Different Noise Distributions

In the previous experiments, noise is injected under a uniform distribution. To further investigate the impact of different noise distributions, we introduce imbalanced noise settings with varying degrees of imbalance. In this setting, a subset of known classes is designated as high-noise classes, while the remaining classes are treated as low-noise classes. When constructing the noisy training data, IND noise is more likely to be drawn from high-noise classes, and OOD samples are more likely to be assigned to these classes, leading to an uneven distribution of noise across classes.

To further control the imbalance strength, we consider three different degrees of noise imbalance, namely mild, moderate, and severe settings. In the mildly imbalanced setting, 60% of IND noise samples are drawn from the high-noise classes and the remaining 40% are drawn from the low-noise classes, while a similar proportion of OOD samples

Table 5
Performance on StackOverflow under different noise distributions and imbalance degrees (25% known class ratio).

OOD Type	Noise Distribution	20% Noise				40% Noise			
		F1-All	ACC	F1-K	F1-U	F1-All	ACC	F1-K	F1-U
Near	Uniform	**77.62**	**78.19**	**76.99**	**80.75**	**80.68**	**82.27**	**79.77**	**85.23**
	Imbalance (Mild)	76.04	77.77	75.50	78.74	79.49	80.95	78.54	84.20
	Imbalance (Moderate)	73.73	73.97	73.70	73.87	77.96	78.82	77.14	82.06
	Imbalance (Severe)	75.40	76.01	75.07	77.00	74.98	79.96	73.49	82.44
Far	Uniform	76.02	76.03	75.59	**78.20**	**79.17**	**80.91**	**78.29**	**83.58**
	Imbalance (Mild)	**76.53**	**77.79**	**76.20**	78.19	78.11	80.75	77.32	82.07
	Imbalance (Moderate)	76.33	76.38	75.99	78.00	75.15	76.45	74.52	78.31
	Imbalance (Severe)	73.38	73.91	73.29	73.87	72.47	74.27	71.62	76.71

Table 6
Open classification performance on real-world noisy datasets with different known class ratios (50%, 75%).

	Datasets	NoisyAG-NewsMed				NoisyAG-NewsWorst			
Known%	Methods	F1-All	ACC	F1-K	F1-U	F1-All	ACC	F1-K	F1-U
50%	DeepUnk [18]	47.70	48.24	62.04	19.04	44.27	47.50	61.46	9.88
	K+1-way [47]	48.49	51.20	49.39	46.68	39.36	44.47	35.24	47.61
	CNOIC (Ours)	**72.98**	**74.20**	**71.13**	**76.69**	**67.80**	**70.37**	**64.29**	**74.82**
75%	DeepUnk [18]	58.00	59.62	72.20	15.40	56.40	59.10	**70.37**	14.48
	K+1-way [47]	22.72	32.28	17.96	36.56	39.14	43.31	39.32	38.58
	CNOIC (Ours)	**72.23**	**70.92**	**76.26**	**60.17**	**64.69**	**63.04**	67.98	**54.81**

are assigned to these classes. In the moderately imbalanced setting, this proportion increases to 70% versus 30%, and in the severely imbalanced setting, it further increases to 80% versus 20%. These three settings correspond to progressively stronger imbalance in the distribution of both IND noise and OOD label assignment across classes, while keeping the overall noise ratio unchanged. The experimental results under different noise distributions and imbalance degrees are reported in Table 5.

From Table 5, we observe that the uniform noise distribution generally yields the best or highly competitive performance across most settings. As the degree of noise imbalance increases, the performance tends to degrade, especially under moderately and severely imbalanced settings. This is because noise becomes unevenly distributed across classes, making the learning process more challenging and leading to biased representations and reduced generalization ability. Nevertheless, the proposed method still maintains relatively stable performance across different noise distributions, demonstrating its robustness to varying noise patterns, even under severely imbalanced settings.

5.2.4. Performance on Real-World Noisy Data

To further evaluate the effectiveness of the proposed method under realistic noise conditions, we conduct additional experiments on two real-world noisy datasets derived from the AG News corpus, following the construction in prior work [10]. Different from synthetic noise injection, these datasets contain naturally occurring annotation inconsistencies. Specifically, NoisyAG-NewsMed has a moderate noise level with an approximate noise ratio of 20%, while NoisyAG-NewsWorst represents a more challenging setting with a higher noise ratio of around 38%.

In our experiments, a subset of the original datasets is sampled to construct the evaluation benchmark, and two known-class ratios (50% and 75%) are considered to simulate different openness levels. The performance is evaluated using standard metrics including F1-All, ACC, F1-K, and F1-U. The experimental results are summarized in Table 6. It can be observed that the performance of our method, CNOIC, decreases as the noise level increases. However, overall, it consistently outperforms the other methods across different settings, demonstrating its robustness under increasing noise conditions.

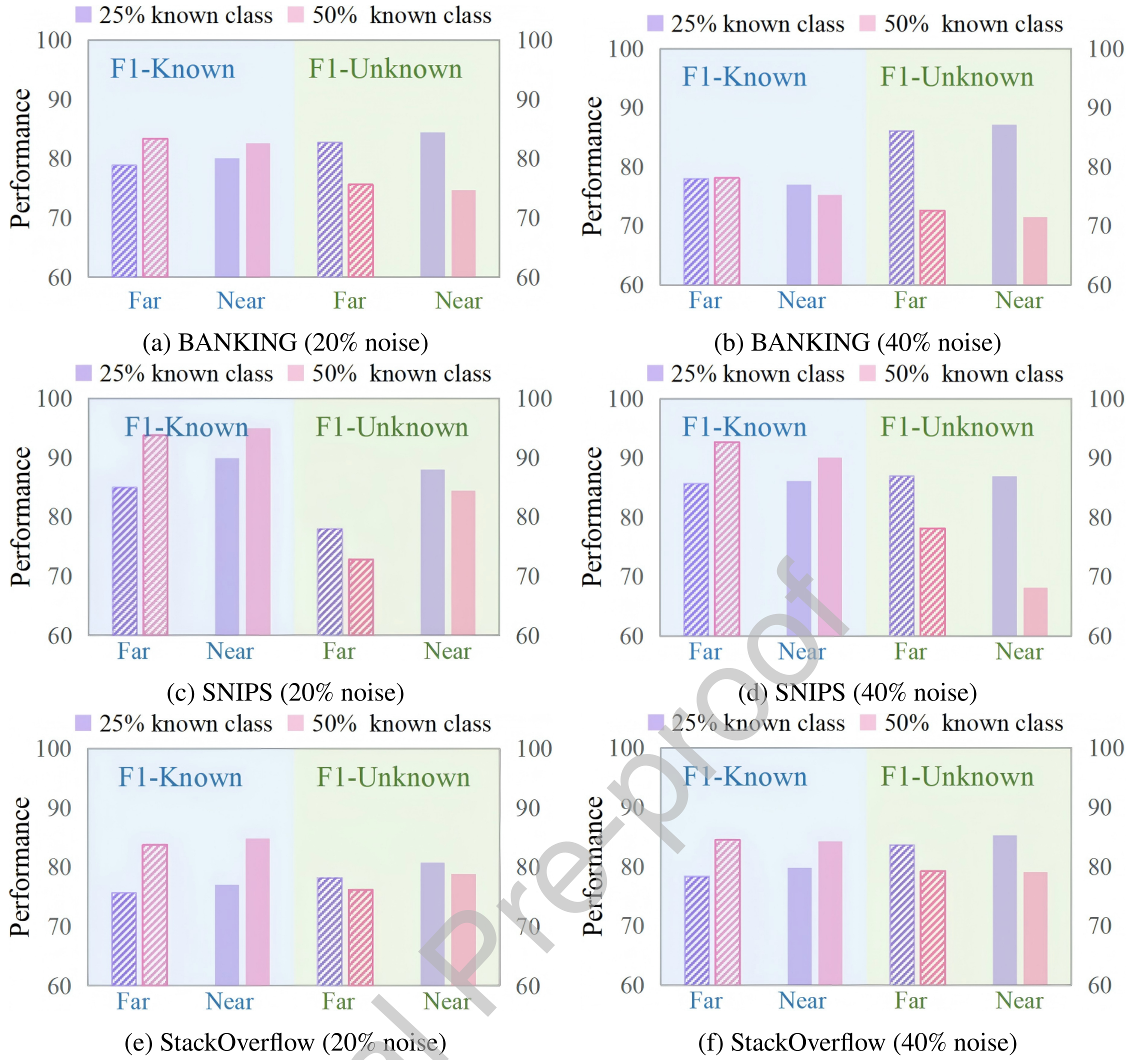


Figure 3: F1-scores for known and unknown classes under varying noise ratios (20%, 40%), known class ratios (25%, 50%), and OOD noise types (Far-OOD vs Near-OOD) on BANKING, SNIPS, and StackOverflow datasets.

5.2.5. Performance on Known and Unknown Class Recognition

To evaluate the performance of the proposed CNOIC method in both known-class classification and unknown-class recognition, we measure the fine-grained F1-scores for known and unknown classes, denoted as F1-Known and F1-Unknown. Fig. 3 presents the results on three benchmark datasets: BANKING, SNIPS, and StackOverflow, under varying noise ratios (20% and 40%), known class proportions (25% and 50%), and OOD noise types (Near and Far).

From Fig. 3, we observe the following key patterns. First, when the known class ratio increases from 25% to 50%, the performance on known-class recognition (F1-Known) consistently improves across all datasets. However, this improvement is accompanied by a noticeable decline in unknown-class recognition (F1-Unknown). This phenomenon indicates a trade-off between fitting known-class distributions and preserving open space for unknown intents. As the model focuses more on known categories, its ability to detect unknown classes becomes relatively weaker. Second, the effect of different OOD noise types varies across datasets, but in most cases Near-OOD noise leads to better unknown-class recognition performance. Since Near-OOD samples are closer to known classes in the feature space, they help simulate ambiguous decision boundary regions during training, which improves the model's sensitivity to open space. In contrast, Far-OOD samples provide stronger semantic separation but contribute less to modeling boundary uncertainty. Third, increasing the overall noise ratio does not always lead to performance degradation for unknown-class detection. In several settings, especially when the known class ratio is low, higher noise levels slightly improve F1-Unknown. This suggests that certain noisy samples, particularly OOD noise, may provide useful signals that help the model better explore open regions in the representation space and improve its ability to identify unseen intents.

5.2.6. *Statistical Significance Analysis*

To assess the statistical significance of the performance differences, we conduct the Friedman test followed by the Nemenyi post-hoc test under the near-OOD setting. Specifically, we evaluate a total of eight methods, including the proposed CNOIC and seven baseline approaches. The analysis is performed on the overall classification accuracy (ACC) across multiple experimental configurations. We consider three benchmark datasets, two known-class ratios, and two noise ratios, resulting in 12 experimental settings (3 datasets $\times$ 2 known-class ratios $\times$ 2 noise ratios). For each setting, the results over five random seeds are averaged to ensure a fair and stable comparison.

The Friedman test yields a statistic of $\chi^2(7) = 55.56$ with a $p < 0.01$, indicating that the performance differences among the compared methods are statistically significant. Fig. 4 presents the Critical Difference (CD) diagram obtained from the Nemenyi test at the significance level of 0.05, where the horizontal axis represents the average rank of each method across all tasks, and a smaller rank indicates better performance.

As shown in Fig. 4, the proposed CNOIC achieves the best average rank among all eight methods. Moreover, the rank differences between CNOIC and most baseline methods, including DETER, Ellipsoid, DeepUnk, OLT, LN-KNN, and GBRAIN, exceed the critical difference threshold, indicating statistically significant improvements. Although the difference between CNOIC and k+1 does not exceed the critical difference, CNOIC still achieves the lowest overall rank. These results demonstrate that the proposed method consistently achieves superior and robust performance under diverse noisy and open-set conditions.

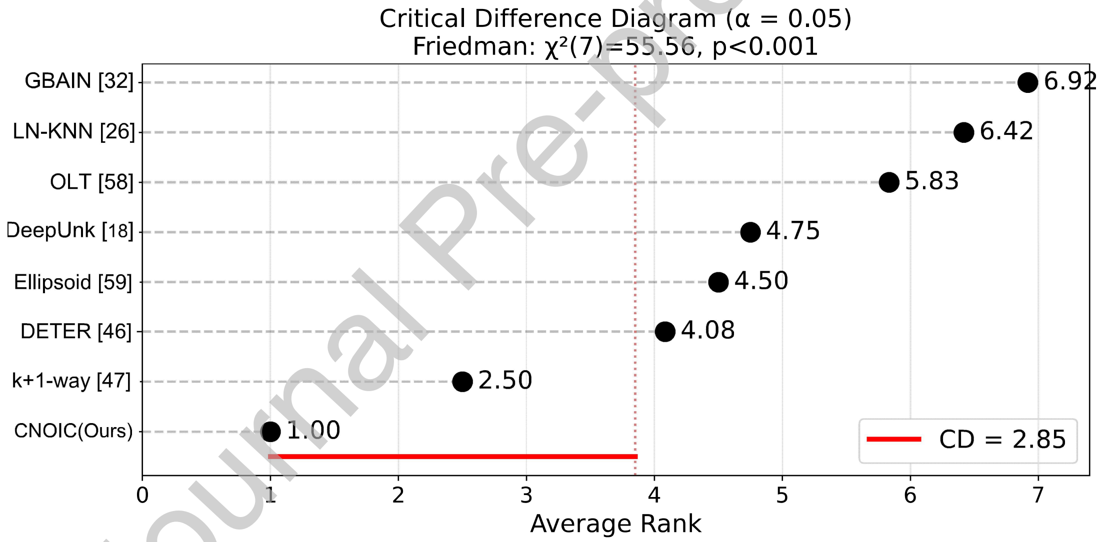
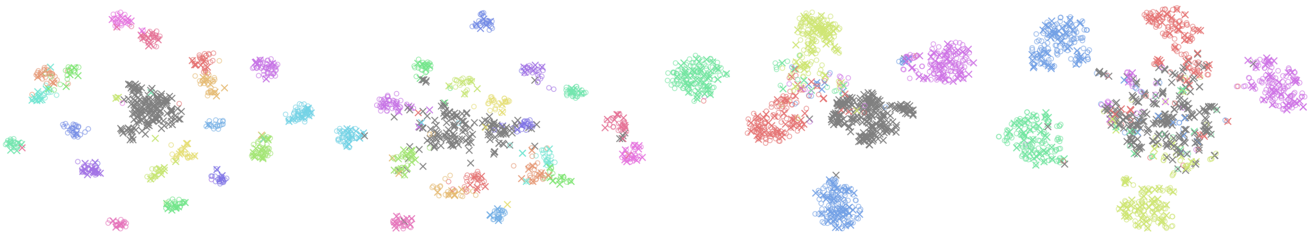


Figure 4: Critical Difference (CD) diagram of the Nemenyi test across the 12 experimental tasks. The red line indicates the CD at the significance level $\alpha = 0.05$.

5.2.7. *t-SNE Visualization Analysis*

Fig. 5 illustrates the feature representations learned by CNOIC, sampled from the training set. Each subfigure includes clean samples (colored circles), 20% IND noise (colored crosses), and 20% OOD noise (gray crosses). Specifically, Fig. 5a and Fig. 5b correspond to the BANKING dataset with Far-OOD and Near-OOD noise, respectively, while subfigures Fig. 5c and Fig. 5d depict the StackOverflow dataset under the same noise conditions. From the visualizations, we observe that most IND noise samples are located within the representation space of their true known classes, whereas OOD noise samples are positioned farther from known class regions. This suggests that CNOIC learns representations in which IND and OOD noise exhibit distinguishable distribution patterns, while encouraging OOD samples to serve as proxies for unknown regions. Additionally, the representations of OOD noise are more compact when the injected noise originates from Far-OOD sources, and more dispersed when derived from Near-OOD sources. This phenomenon arises because Far-OOD samples are semantically more distant from known classes compared to



(a) BANKING (Far) (b) BANKING (Near) (c) StackOverflow (Far) (d) StackOverflow(Near)

Figure 5: The t-SNE visualization of embedded features in the training set with 40% noise. Different colors represent different classes. Colored circles denote clean samples, colored crosses indicate IND noise, and grey crosses denote OOD noise.

Near-OOD samples, which share partial similarities. These findings demonstrate that CNOIC captures meaningful semantic structures within the representation space.

5.3. Ablation Study

To further investigate the effectiveness of each component in the proposed framework, we conduct ablation experiments by removing different modules or loss functions. Specifically, we evaluate the following variants:

- **w/o GB**: The granular-ball based noise identification module is removed, while the multi-granularity decision boundary used in inference is retained.
- **w/o $\mathcal{L}_{\text{intra}}$**: The intra-class compactness loss is removed, so prototypes of the same class are no longer explicitly encouraged to stay close in the representation space.
- **w/o $\mathcal{L}_{\text{inter}}$**: The inter-class separation loss is removed, which eliminates the constraint that pushes prototypes of different classes apart.
- **w/o $\mathcal{L}_{\text{OOD}}$**: The OOD separation loss is removed, meaning that known-class prototypes are no longer explicitly separated from OOD prototypes.

Table 7 reports the ablation results on the StackOverflow dataset under the near-OOD setting with a 25% known-class ratio. All results are averaged over five random seeds. The full model consistently achieves the best performance under both noise settings, demonstrating the effectiveness of the proposed framework. When the granular-ball based noise identification module is removed (w/o GB), the performance drops noticeably, especially under the 40% noise setting. This indicates that the proposed granular-ball structure plays a critical role in distinguishing clean samples from noisy ones and becomes increasingly important as the noise level grows.

The loss-level ablation further confirms the contribution of each structural constraint. Removing $\mathcal{L}_{\text{intra}}$ slightly degrades performance, suggesting that encouraging prototypes of the same class to remain compact helps preserve class-wise semantic structure. Removing $\mathcal{L}_{\text{inter}}$ leads to a larger performance drop, indicating that maintaining sufficient separation between class prototypes is important for discriminative representation learning. In addition, removing $\mathcal{L}_{\text{OOD}}$ also reduces performance, demonstrating that explicitly separating known prototypes from OOD prototypes helps reserve open space for potential unknown intents. Overall, these results verify that both the granular-ball noise identification mechanism and the proposed loss design jointly contribute to robust open intent classification under complex noisy conditions.

5.4. Parameter Study

We analyze the impact of several key thresholds in our CNOIC framework, including: (1) the purity threshold p_l and size threshold n_l used in granular-ball clustering in Sec. 4.2; (2) the high- and low-quality purity thresholds p_h and p_w for noise type recognition in Sec. 4.3; and (3) the purity threshold p_t and size threshold n_t used in the inference stage for open intent classification in Sec. 4.5.

Table 7
Ablation study results on StackOverflow (25% known class ratio).

Method	20% Noise				40% Noise			
	F1-All	ACC	F1-K	F1-U	F1-All	ACC	F1-K	F1-U
CNOIC (Ours)	**77.62**	**78.19**	**76.99**	**80.75**	**80.68**	**82.27**	**79.77**	**85.23**
w/o GB	76.62	76.10	76.29	78.25	75.43	77.13	74.48	80.19
w/o $\mathcal{L}_{\text{intra}}$	76.22	76.78	75.64	79.11	77.74	79.55	76.79	82.49
w/o $\mathcal{L}_{\text{inter}}$	75.31	74.92	74.96	77.10	78.20	79.57	77.34	82.50
w/o $\mathcal{L}_{\text{OOD}}$	76.63	77.16	76.16	79.00	77.02	78.23	76.23	81.00

Table 8
Effect of the purity threshold p_l on granular-ball structure (with $n_l = 12$ fixed).

p_l	#Balls	Number of granular-balls per class					Purity distribution				
		label_0	label_1	label_2	label_3	label_4	[0.0,0.3)	[0.3,0.5)	[0.5,0.7)	[0.7,0.9)	[0.9,1.0]
0.60	65	11	11	15	9	19	2	20	24	15	4
0.65	65	11	11	15	9	19	2	20	24	15	4
0.70	66	11	11	16	9	19	2	20	24	16	4
0.75	66	11	11	16	9	19	2	20	24	16	4
0.80	68	11	11	18	9	19	2	20	25	16	5
0.85	68	11	11	18	9	19	2	20	25	15	6

Table 9
Effect of the minimum ball size threshold n_l on granular-ball structure (with $p_l = 0.8$ fixed).

n_l	#Balls	Number of granular-balls per class					Purity distribution				
		label_0	label_1	label_2	label_3	label_4	[0.0,0.3)	[0.3,0.5)	[0.5,0.7)	[0.7,0.9)	[0.9,1.0]
7	74	11	12	18	9	24	3	14	32	19	6
9	72	11	13	18	9	21	3	19	27	17	6
11	69	11	11	18	9	20	3	19	26	16	5
13	67	11	11	18	9	18	2	19	25	16	5
15	61	11	9	14	9	18	2	20	22	13	4

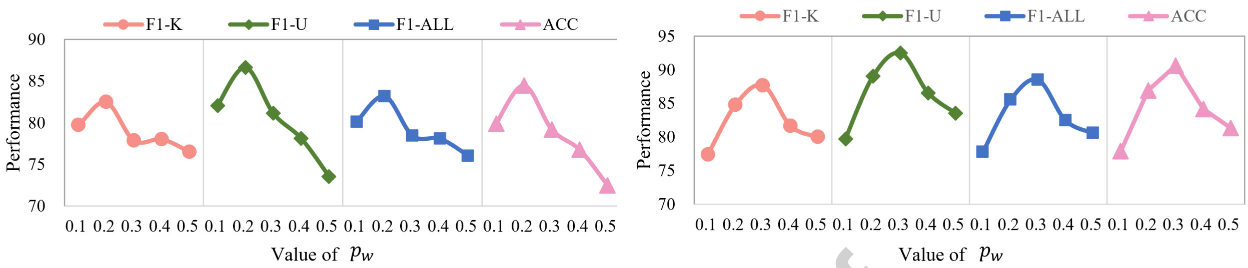
5.4.1. *Parameter Sensitivity Analysis of p_l and n_l*

During granular-ball construction in training, a granular-ball will be further split when its purity is lower than p_l and its size is larger than n_l, generating new granular-balls. In general, a larger p_l and a smaller n_l correspond to stricter splitting conditions, which tend to produce more granular-balls. To investigate the influence of p_l and n_l on the resulting granular-ball structure, we collect the statistics of granular-balls under different parameter settings from a representative clustering process during training, including the total number of generated granular-balls, the number of granular-balls for each class, and the purity distribution. These results are obtained on the StackOverflow dataset with a 25% known class ratio and 20% total noise (10% IND noise and 10% Near-OOD noise), as reported in Tables 8 and 9.

Table 8 shows the variation of the granular-ball structure when n_l is fixed at 12 and p_l increases from 0.60 to 0.85. Overall, as the purity requirement becomes stricter, the total number of generated granular-balls increases slightly (from 65 to 68), accompanied by a corresponding growth in high-purity granular-balls. Meanwhile, both the class-wise distribution of granular-balls (e.g., label_0, label_1, label_3, and label_4) and the purity distribution remain largely unchanged across different settings. Table 9 presents the variation of the resulting granular-ball structure when p_l is fixed at 0.8 and n_l increases from 7 to 15. As n_l increases, the total number of granular-balls gradually decreases, indicating that the partition becomes coarser under a larger minimum ball size constraint. In this case, the number of granular-balls for certain classes (e.g., label_0 and label_3) remains nearly unchanged, while those for other classes (e.g., $label_1$, $label_2$, and $label_4$) exhibit slight decreases. In terms of purity distribution, as n_l increases, the number of low-purity granular-balls (e.g., [0.3,0.5)) shows a slight increasing trend, whereas higher-purity granular-balls (e.g., [0.7,0.9)) decrease moderately.

Overall, the results reveal a consistent pattern: the granular-ball structure changes smoothly with respect to both p_l and n_l. In particular, when the parameter variations are within a moderate range, their influence on the number and distribution of granular-balls remains limited, without causing significant structural changes.

5.4.2. Parameter Sensitivity Analysis of p_h and p_w



(a) p_h fixed as 0.8 (b) p_h fixed as 0.9

Figure 6: Model performance on the StackOverflow dataset as p_w varies, with a fixed p_h value of 0.8 in (a) and 0.9 in (b).

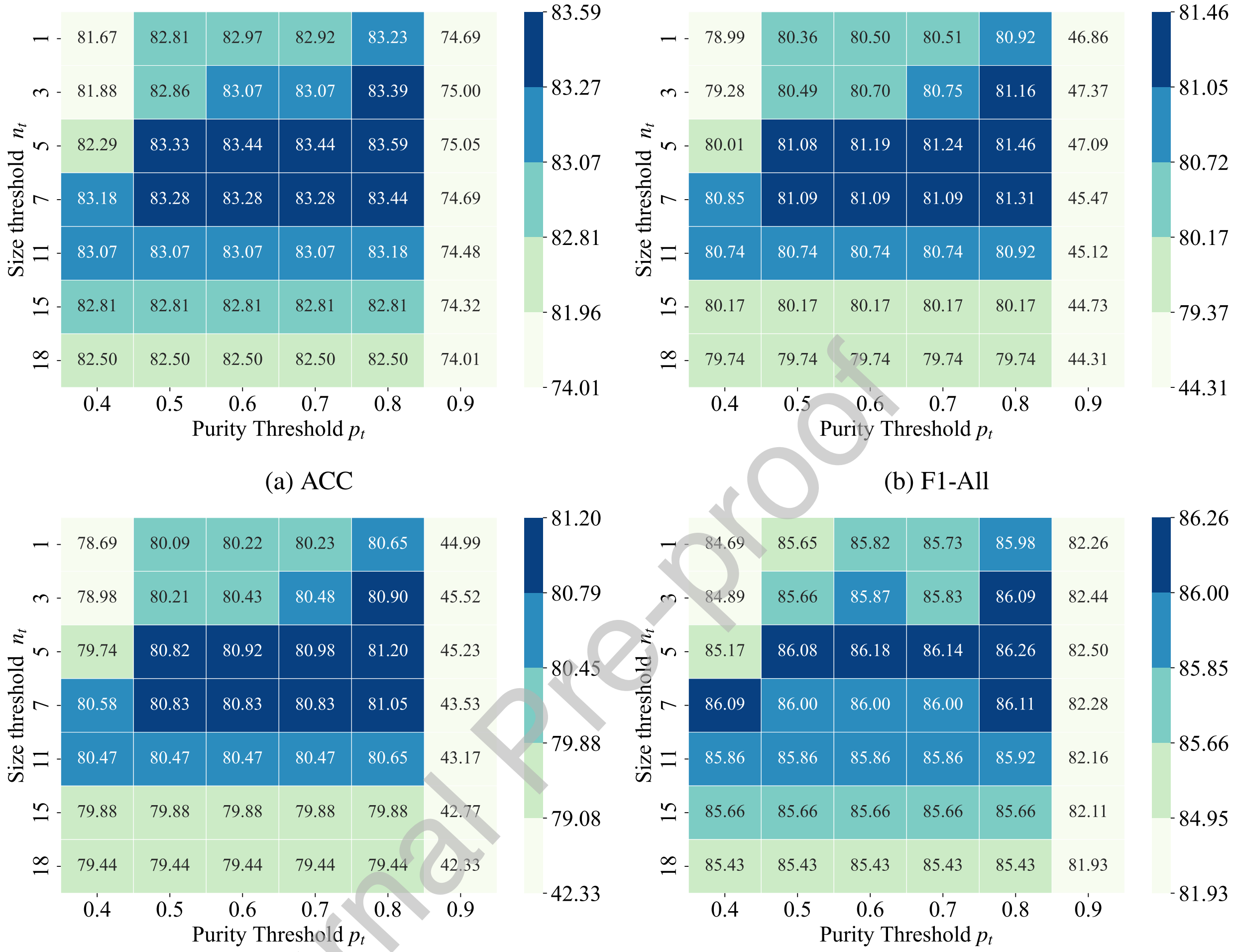
We investigate the impact of the purity thresholds p_h and p_w, which are used to distinguish granular-balls of different quality levels during noise type recognition. Specifically, p_h serves as the threshold for identifying high-quality granular-balls with strong label consistency, while p_w is used to detect low-quality granular-balls that are more likely to contain OOD noise. Fig. 6 presents the experimental results on the StackOverflow dataset under the setting of a 25% known class ratio, with 10% IND noise and 10% Near-OOD noise. Specifically, we evaluate the model's performance using four metrics: F1-K, F1-U, F1-All, and ACC. We vary p_w from 0.1 to 0.5 in increments of 0.1, under two fixed values of p_h, namely $p_h = 0.8$ and $p_h = 0.9$, as shown in Fig. 6a and Fig. 6b.

It can be observed that the model achieves its best performance at $p_w = 0.2$ when $p_h = 0.8$, and at $p_w = 0.3$ when $p_h = 0.9$. Overall, we observe a consistent trend where model performance first increases and then decreases as p_w grows. This suggests that a moderate increase in p_w (e.g., to 0.2 or 0.3) helps the model better identify low-purity granular-balls associated with OOD noise, thereby improving open intent classification through the open space-aware representation learning module. However, an overly high p_w may cause clean or IND noise samples to be mistakenly treated as OOD noise, causing their representations to deviate from their true classes in training and resulting in performance degradation.

5.4.3. Parameter Sensitivity Analysis of p_t and n_t.

In the inference stage, we construct multi-granularity decision boundaries for open intent classification by selecting granular-balls whose purity is higher than the threshold p_t and whose size is larger than n_t. The purity threshold p_t controls the reliability of the selected granular-balls, while the size threshold n_t determines the minimum number of samples required for each granular-ball. Together, these two parameters regulate the trade-off between representation reliability and the coverage of known-class regions. To investigate the joint influence of p_t and n_t, we conduct a parameter sensitivity analysis on the BANKING dataset under the setting of a 25% known class ratio with 10% IND noise and 10% Near-OOD noise. Specifically, p_t varies from 0.4 to 0.9 and n_t ranges from 1 to 18. The corresponding performance variations are illustrated in Fig. 7, where four evaluation metrics, including ACC, F1-All, F1-K, and F1-U, are reported.

From Fig. 7, several observations can be drawn. First, when the purity threshold p_t is relatively low, the model tends to include more granular-balls for boundary construction, which increases the coverage of known-class regions but may introduce unreliable representatives. As p_t increases to a moderate range (e.g., $p_t \in [0.6, 0.8]$), the performance of all metrics becomes more stable and generally reaches higher values, indicating that moderately strict purity constraints help select more reliable granular-ball representatives. Second, the size threshold n_t mainly controls the robustness of granular-ball selection. When n_t is small, more granular-balls are included, which increases boundary flexibility but may introduce noisy structures. As n_t increases, the number of selected granular-balls decreases, leading to more stable but slightly less flexible decision boundaries. The results show that the performance remains relatively stable within a moderate range of n_t, demonstrating the robustness of the proposed method to this parameter. Overall, the proposed method exhibits stable performance across a wide range of parameter settings, indicating that the multi-granularity boundary construction mechanism is not overly sensitive to the choice of p_t and n_t. In our experiments, the best performance is generally achieved when p_t is around 0.7–0.8 and n_t lies within a moderate range, which balances the reliability and coverage of granular-ball representations.



(a) ACC

(b) F1-All

(c) F1-K

(d) F1-U

Figure 7: Parameter sensitivity analysis with respect to the purity threshold p_t and size threshold n_t.

5.5. Computational Complexity and Efficiency Evaluation

We analyze both the theoretical complexity and empirical efficiency of the proposed method.

The main computational cost of the proposed method arises from the granular-ball construction and subsequent representation learning process. Given N samples with feature dimension d, the initialization stage computes distances from all samples to the initial center once, resulting in a complexity of $O(Nd)$. For a granular-ball containing n samples, one splitting step mainly involves purity evaluation and sample reassignment based on distance comparison, with a complexity of $O(nd)$. Since only low-purity granular-balls are further split, and the splitting process is controlled by the purity threshold and minimum ball size, the number of iterations is limited. Let L denote the maximum splitting depth. Then, the overall complexity of granular-ball construction can be expressed as $O(NdL)$. In practice, as d is fixed and L is typically small, the granular-ball construction exhibits approximately linear complexity with respect to the number of samples. The subsequent training process follows standard mini-batch optimization and does not introduce additional significant overhead compared with existing deep learning-based methods.

To further evaluate the computational efficiency of the proposed method, we compare the runtime and peak memory consumption of different approaches on the StackOverflow dataset under various settings. The results are

Table 10
Comparison of runtime and peak memory on the StackOverflow dataset.

known%	noise%	OOD type	Time (Min)				Peak Memory (GB)			
			OLT [58]	DETER [46]	GBRAIN [32]	CNOIC (Ours)	OLT [58]	DETER [46]	GBRAIN [32]	CNOIC (Ours)
0.25	20%	Near	1.90	3.87	4.17	2.10	5.44	8.82	9.21	10.14
		Far	2.09	4.49	3.98	1.91	5.44	8.82	9.21	10.26
	40%	Near	1.85	4.24	4.16	3.30	5.44	8.82	9.21	7.41
		Far	2.20	4.47	4.10	2.40	5.44	8.82	9.21	7.45
0.5	20%	Near	3.53	6.81	6.19	5.26	5.47	8.82	9.21	9.96
		Far	3.60	6.45	6.26	3.86	5.45	8.82	9.21	9.99
	40%	Near	3.56	6.76	6.20	5.30	5.46	8.82	9.21	7.38
		Far	3.46	6.78	6.32	5.02	5.45	8.82	9.21	7.48

shown in Table 10. The known-class ratios are set to 25% and 50%, the noise ratios are 20% and 40%, and both near and far OOD noise are considered. All experiments were conducted on an NVIDIA A100 80GB GPU with an Intel Xeon Gold 6338 CPU (16 cores) and CUDA 12.8. From the runtime perspective, CNOIC demonstrates competitive efficiency. Specifically, it consistently requires less runtime than DETER and GBRAIN across all settings, while only slightly higher than OLT, which adopts a simpler framework. This indicates that the proposed granular-ball based noise handling mechanism does not introduce significant computational overhead. In terms of Peak memory consumption, CNOIC requires slightly more memory than some lightweight baselines due to the additional granular-ball structures used for noise identification and representation learning. Nevertheless, the memory usage remains comparable to other structure-based methods and stays within a reasonable range. Overall, these results show that CNOIC achieves a good balance between computational efficiency and modeling capability.

6. Conclusions

This work presents a novel attempt to address open intent classification under complex-noise conditions involving both IND and OOD noise. First, the proposed structural knowledge representation module captures rich and robust information of known classes, effectively mitigating the impact of noise. Based on this knowledge, a structural knowledge-guided noise recognition strategy is developed to accurately identify and differentiate various types of noise. Furthermore, an open space-aware representation learning module is introduced to learn discriminative representations that preserve space for potential unknown classes. Finally, multi-granularity decision boundaries are constructed for reliable open intent classification. Extensive experiments across multiple datasets demonstrate the superior performance and robustness of our method compared to existing approaches.

Despite these promising results, the proposed framework involves a multi-stage design with several components, which introduces additional computational complexity compared with simpler baseline methods. While this design is necessary to address the challenges of complex noise and open-set scenarios, improving the efficiency and scalability of the framework remains an important direction for future work. In the future, we will further explore how to enhance the reliability of noise identification under more diverse and dynamic noise patterns. Moreover, how to simplify the model architecture while maintaining performance, and how to extend CNOIC to support continual open intent discovery, are also promising directions for future research.

Acknowledgments

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Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.